AVOID FIRE ACCIDENTS IN ELECTRIC VEHICLE WITH MULTI FAULT DETECTION USING AI AND IOT

A PROJECT REPORT

Submitted by

BOOMIKA P **613521105007**

KAVINAYA P **613521105020**

SUVITHA M **613521105045**

In partial fulfillment for the award of the degree

of

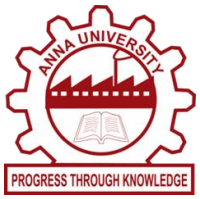
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ANNA UNIVERSITY :: CHENNAI 600 025

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Submitted by

BOOMIKA P **613521105007**

KAVINAYA P **613521105004**

SUVITHA M **613521105017**

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BONAFIDE CERTIFICATE

Certified that this project report **"AVOID FIRE ACCIDENTS IN ELECTRIC VEHICLE WITH MULTI FAULT DETECTION USING AI AND IOT"** is the bonafide work of **"BOOMIKA P (613521105007), KAVINAYA P (613521105020), SUVITHA M (613521105045)"** who carried out the project work under my supervision**.**

SIGNATURE

Dr.S.SENTHILKUMAR.,

HEAD OF THE DEPARTMENT

Professor (CAS),

Electrical And Electronics Engineering,

Government College of Engineering,

Dharmapuri-636704.

SIGNATURE

Dr.P.KARPAGAVALLI.,

SUPERVISOR

Associate Professor (CAS),

Electrical And Electronics Engineering,

Government College of Engineering,

Dharmapuri-636704.

Submitted for the University Examination held at Government College of Engineering, Dharmapuri on ……

INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

We affirm that the project work titled **"AVOID FIRE ACCIDENTS IN ELECTRIC VEHICLE WITH MULTI FAULT DETECTION USING AI AND IOT"** being submitted in partial fulfillment for the award of **BACHELOR OF ENGINEERING** is the original work carried out by us. It has not been part of any other project work submitted for an award of degree either in this or any other university.

Signature of the candidates

BOOMIKA P **KAVINAYA P**

613521105007 **613521105020**

SUVITHA M

613521105045

I Certify that the declaration made above by the candidates is true .

Place:

Date:

Dr. P. KARPAGAVALLI,

Associate Professor,

Electrical and Electronics Engineering,

Government college of Engineering ,

Dharmapuri-636704.

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ABSTRACT

Battery storage forms the most important part of any electric vehicle (EV) as it stores the necessary energy for the operation of EV. So, in order to extract the maximum o/p of a battery & to ensure its Power transfer it is necessary that a efficient battery management system exist is the same. It monitors the Parameters, determine Hardware and software and provide necessary services to ensure safe operation of battery. The proposed system only monitor the battery and Power transfer but also protect it to avoid accidents from occurring. In the focus is on leveraging On-board Diagnostics (OBD) technology, which is a self-diagnostic system in vehicles, to enhance the monitoring and reporting of Electric Vehicle (EV) battery conditions. The primary goal is to detect and analyze the state of EV batteries and provide real-time alerts to vehicle users via their mobile phones. To achieve this, the project proposes the implementation of Deep learning-based data analysis techniques for predicting battery conditions. This involves collecting data from the vehicle's OBD system, such as voltage levels, temperature, charging cycles, etc., and using Deep learning algorithms to analyze this data and predict the current and future health status of the EV battery. The aims to introduce a new design for battery monitoring during charging. This could involve the development of specialized hardware or software solutions that enhance the monitoring capabilities of the EV battery during the charging process, ensuring optimal charging conditions and prolonging battery life.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
EV	ELECTRIC VEHICLE
BMS	BATTERY MANAGEMENT SYSTEM
OBD	ON-BOARD DIAGNOSTICS
IOT	INTERNET OF THINGS
LCD	LIQUID CRYSTAL DISPLAY
ADC	ANALOG TO DIGITAL CONVERTER
SOC	STATE OF CHARGE
BLDC	BRUSHLESS DIRECT CURRENT (MOTOR)
GSM	GLOBAL SYSTEM FOR MOBILE COMMUNICATION
VFD	VARIABLE FREQUENCY DRIVE
FPBN	FUZZY PROBABILITY BAYESIAN NETWORK
ICS	INDUSTRIAL CONTROL SYSTEMS
USB	UNIVERSAL SERIAL BUS
DC	DIRECT CURRENT
EEPROM	ELECTRICALLY ERASABLE PROGRAMMABLE READ-ONLY MEMORY